

* 17 : 40

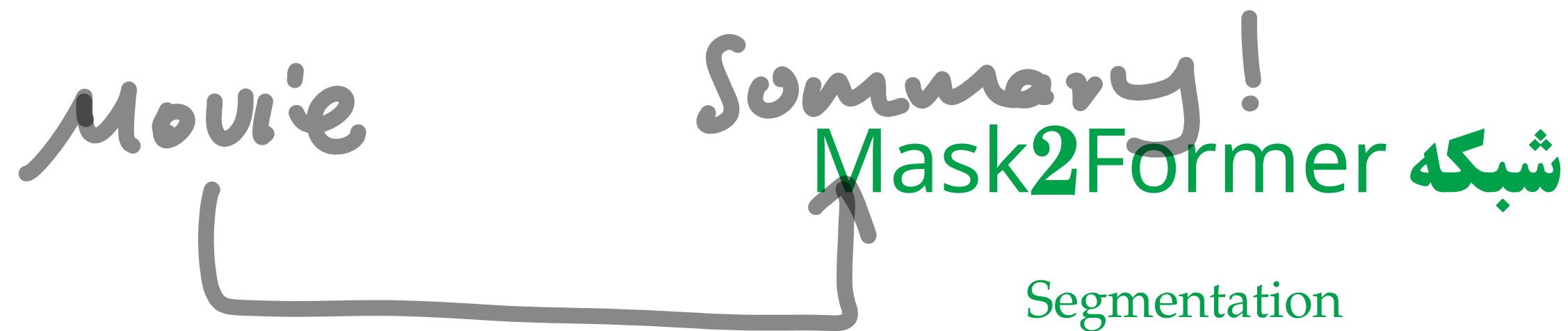
/ CNN ~> Transfer

Mask2Former شبکه

Segmentation



✓ long-term memory!



LM → Transformer based
Image/video to text → TF

5 years ago! →

Image Captioning! → { CNN ✓
+ LSTM ✓

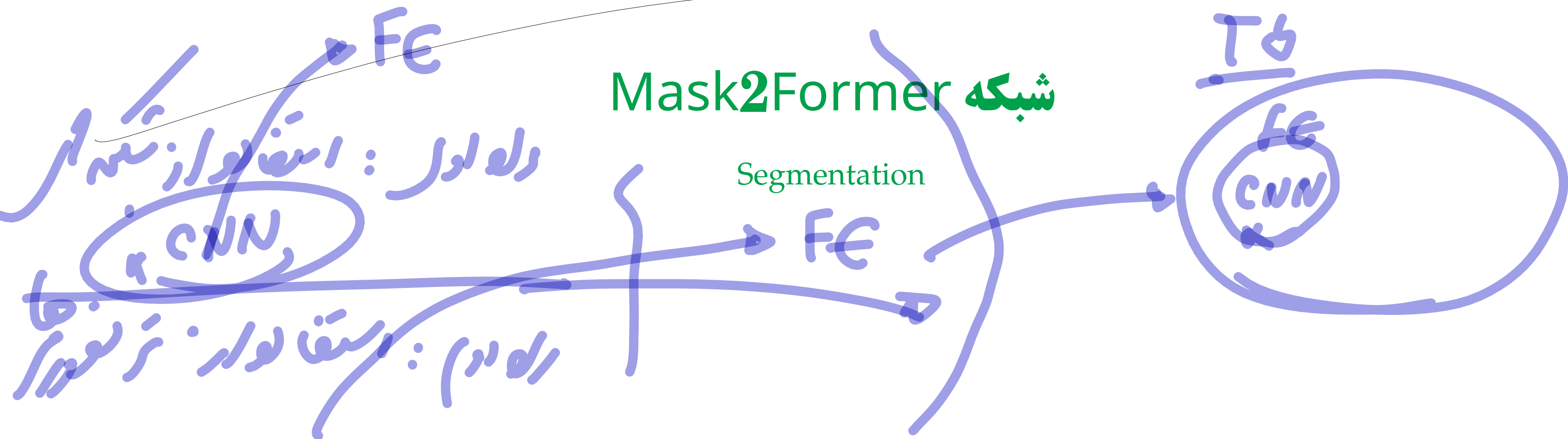
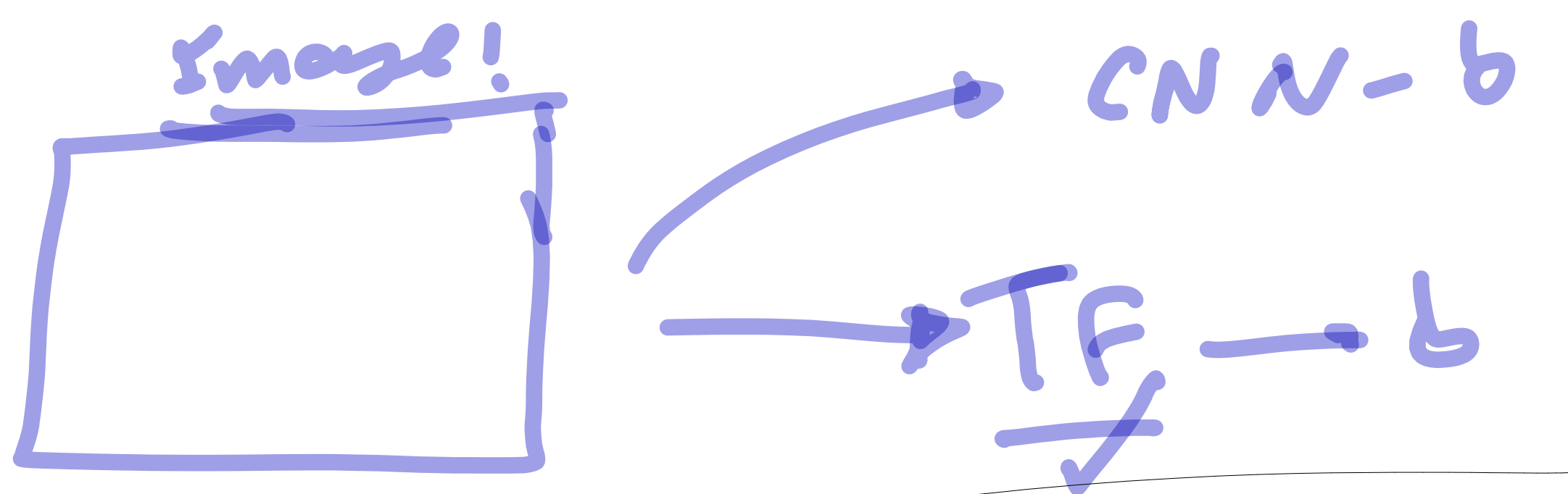
Mask2Former شبکه

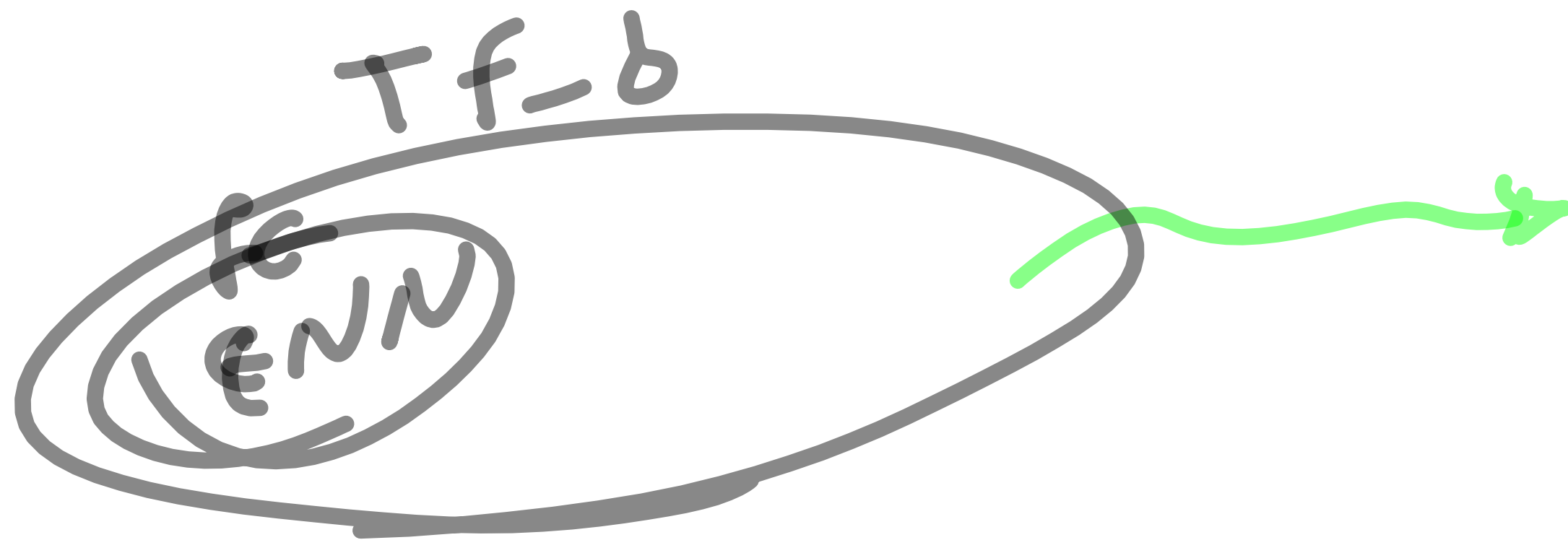
Segmentation

✓ LM → Tr →

vision! → Transf. based!

a dog on the
beach!





Real time :

CNN : mobile
Net

Mask2Former شبکه

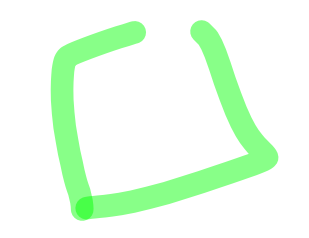
Segmentation

Not / Real

101
CNN : Resnet

segmentation ~ →

CNN



TF

LLM
SLM

BERT

Mask2Former شبکه

Segmentation

TF

CNN

vs

TF-based

FFN

Masked-attention Mask Transformer for Universal Image Segmentation

Bowen Cheng^{1,2*} Ishan Misra¹ Alexander G. Schwing² Alexander Kirillov¹ Rohit Girdhar¹

¹Facebook AI Research (FAIR) ²University of Illinois at Urbana-Champaign (UIUC)

<https://bowenc0221.github.io/mask2former>

Masked-attention Mask Transformer for Universal Image Segmentation

Bowen Cheng^{1,2*} Ishan Misra¹ Alexander G. Schwing² Alexander Kirillov¹ Rohit Girdhar¹

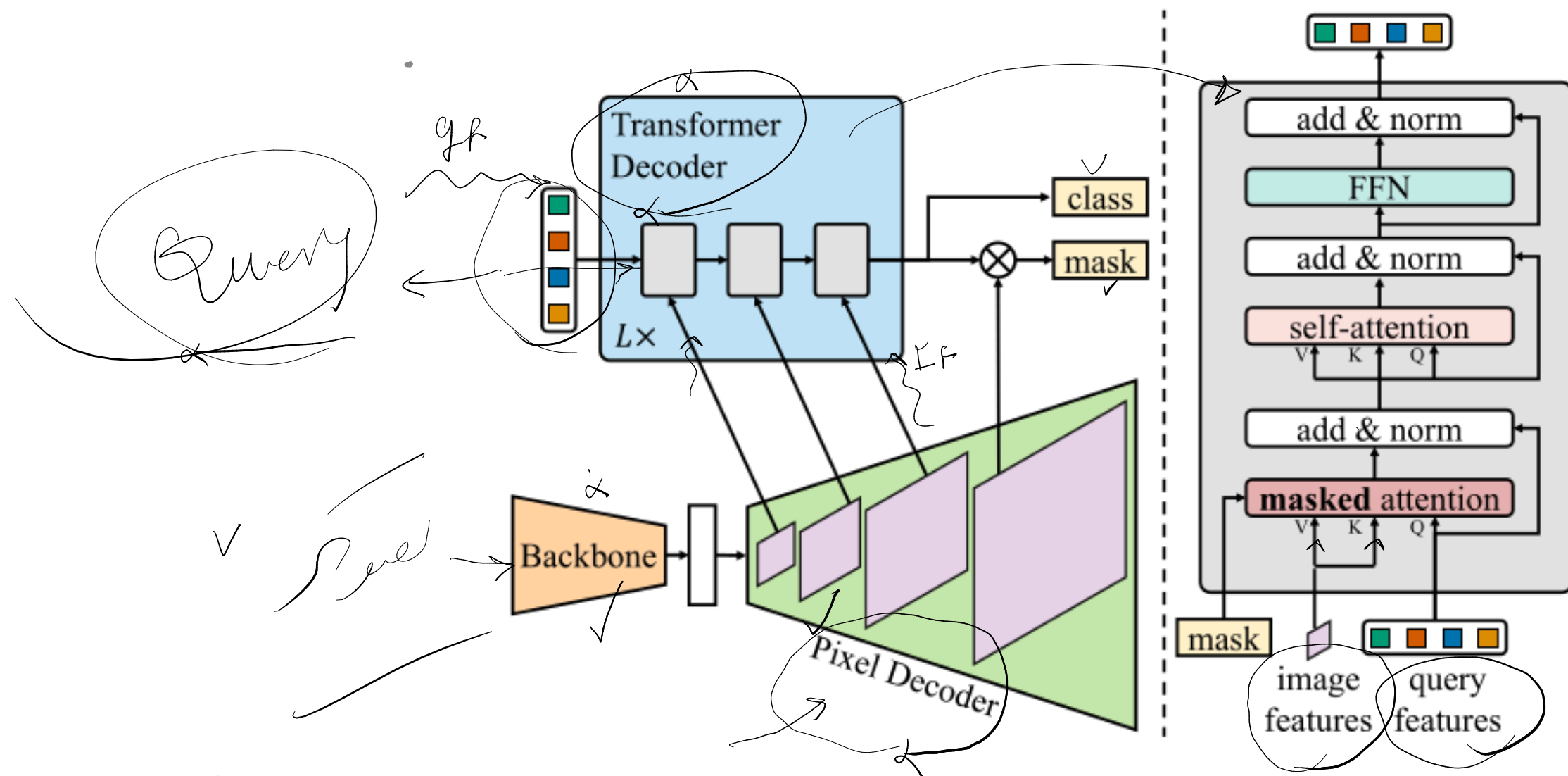
¹Facebook AI Research (FAIR) ²University of Illinois at Urbana-Champaign (UIUC)

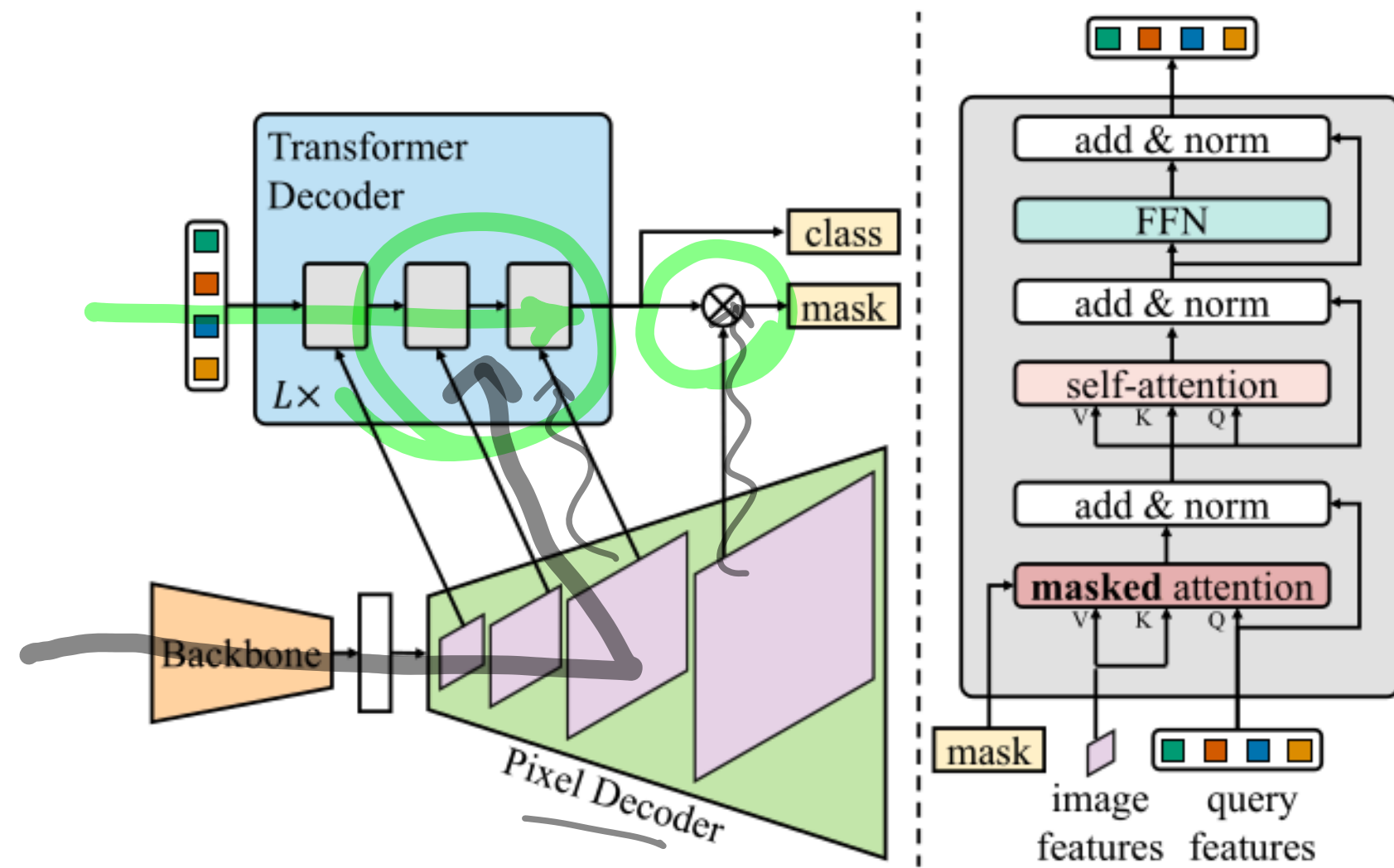
<https://bowenc0221.github.io/mask2former>

①
The
SegFormer

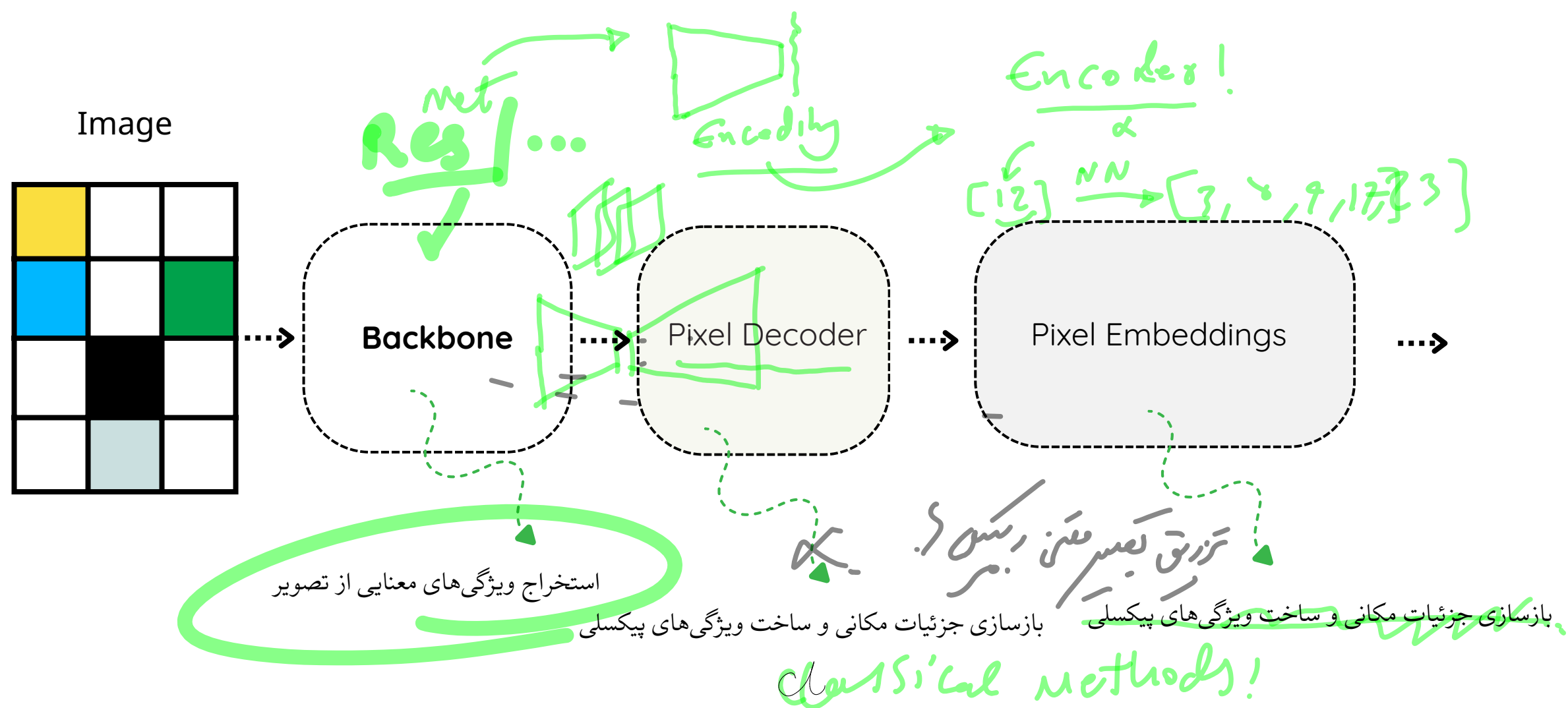
conv
conv
conv

*

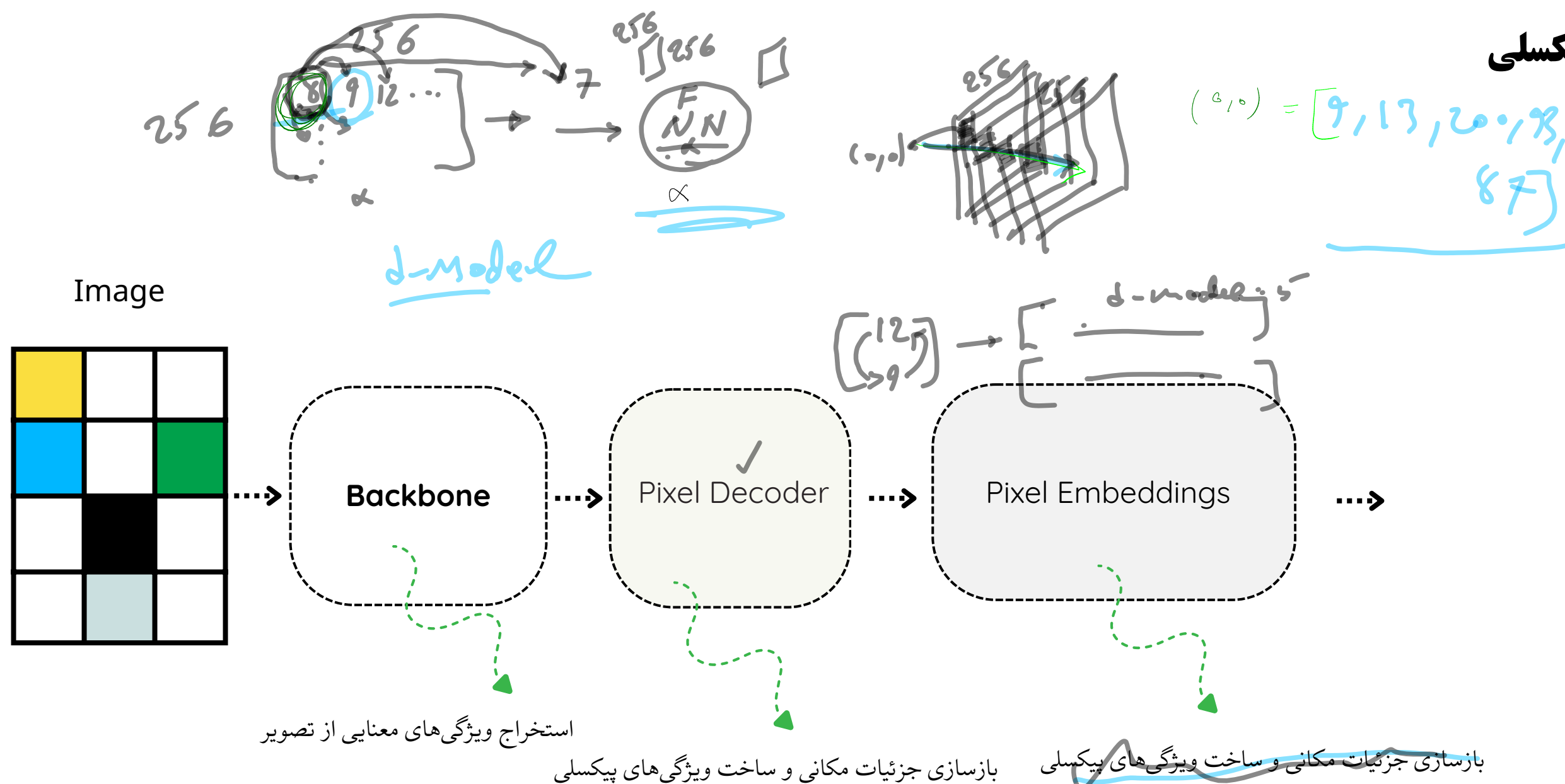




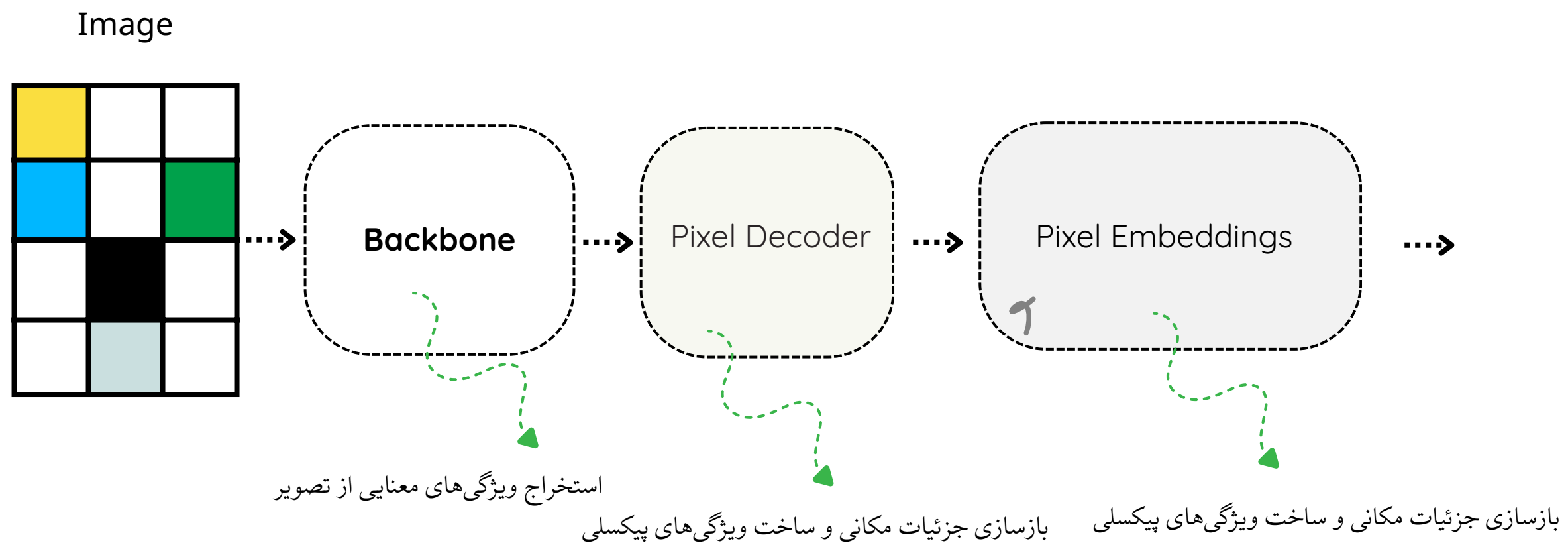
جریان اول : جریان پیکسلی



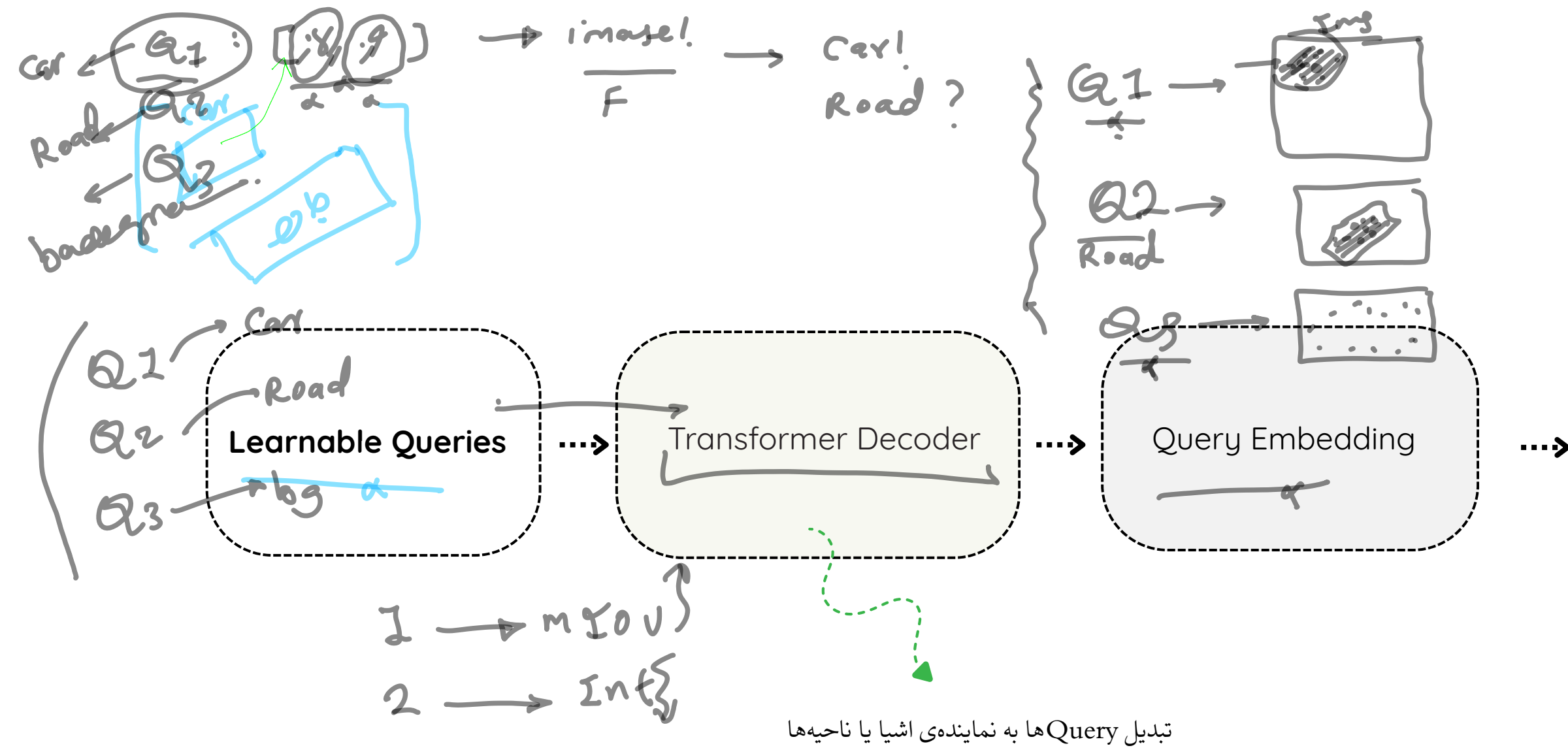
جریان اول : جریان پیکسلی



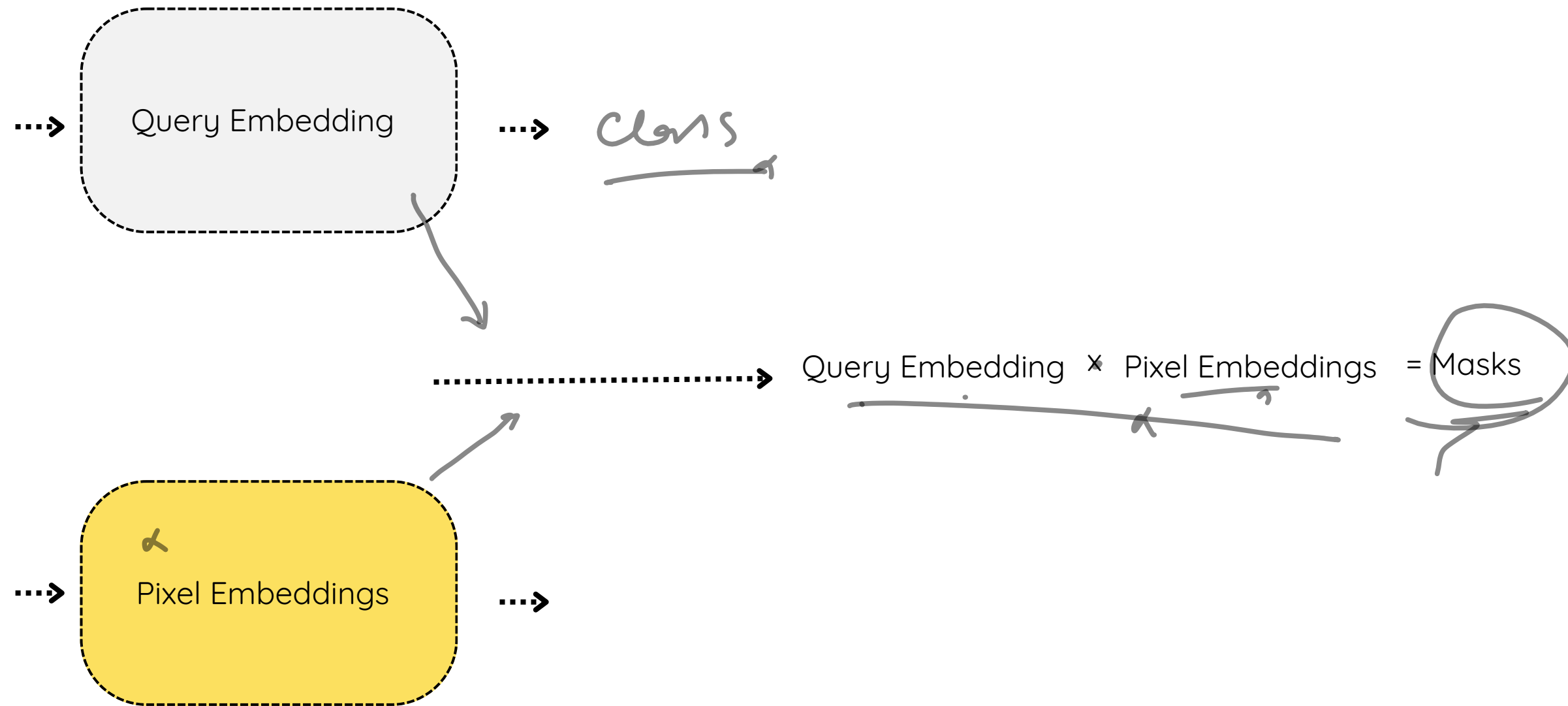
جریان اول : جریان پیکسلی



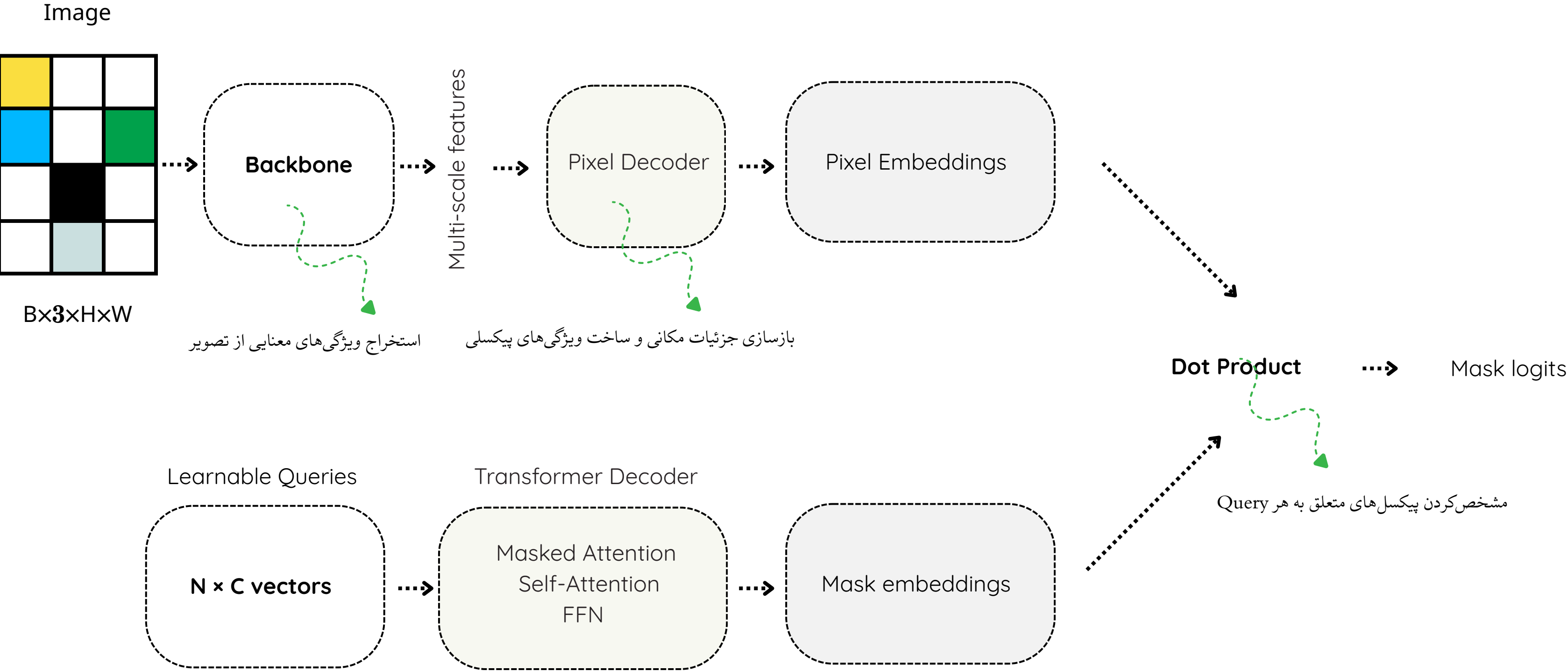
جریان دوم : جریان Query



ادغام دو خروجی



نمای شماتیک کامل



method	backbone	crop size	mIoU (s.s.)	mIoU (m.s.)
MaskFormer [14]	R50	512	44.5	46.7
Mask2Former (ours)	R50	512	47.2	49.2
Swin-UperNet [36, 58]	Swin-T	512	46.1	46.1
MaskFormer [14]	Swin-T	512	46.7	48.8
Mask2Former (ours)	Swin-T	512	47.7	49.6
MaskFormer [14]	Swin-L [†]	640	54.1	55.6
FaPN-MaskFormer [14, 39]	Swin-L-FaPN [†]	640	55.2	56.7
BEiT-UperNet [2, 58]	BEiT-L [†]	640	-	57.0
Mask2Former (ours)	Swin-L [†]	640	56.1	57.3
	Swin-L-FaPN [†]	640	56.4	57.7

Table 3. **Semantic segmentation on ADE20K val with 150 categories.** Mask2Former consistently outperforms MaskFormer [14] by a large margin with different backbones (all Mask2Former models use MSDeformAttn [66] as pixel decoder, except Swin-L-FaPN uses FaPN [39]). Our best model outperforms the best specialized model, BEiT [2]. We report both single-scale (s.s.) and multi-scale (m.s.) inference results. Backbones pre-trained on ImageNet-22K are marked with [†].

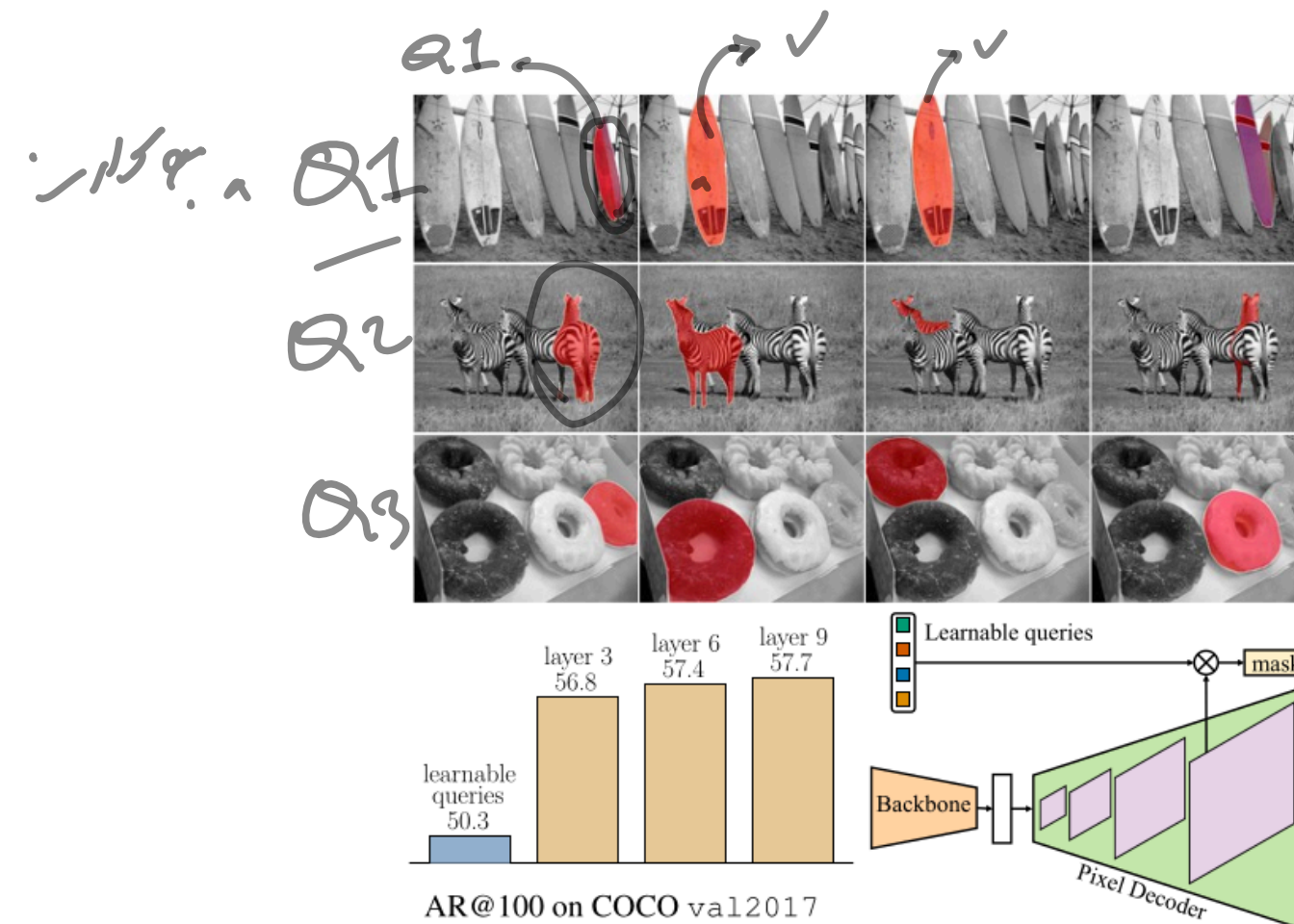


Figure 3. **Learnable queries as “region proposals”.** *Top:* We visualize mask predictions of four selected learnable queries *before* feeding them into the Transformer decoder (using R50 backbone). *Bottom left:* We calculate the class-agnostic average recall with 100 proposals (AR@100) and observe that these learnable queries provide good proposals compared to the final predictions of Mask2Former after the Transformer decoder layers (layer 9). *Bottom right:* Illustration of proposal generation process.